VULNERABLITY ASSESSMENT OF METASPLOITABLE 2 USING GREENBONE/OPENVAS

Prepared For:

Cybersecurity Portfolio Project

Prepared by:

Thato Montjane

Contents

Vulnerability Assessment Report 2

Executive Summary 2

Purpose 2

Scope 2

Assessment Objectives 3

Lab Environment 3

Methodology 3

Phase 1 – Environment Setup 3

Phase 2 – Target Configuration 3

Phase 3 – Vulnerability Scanning 4

Phase 4 – Risk Assessment 4

Executive Findings 4

Critical Findings 4

Finding 1 – TWiki Remote Command Execution 4

Description 5

Risk 5

Recommendation 5

Finding 2 – rlogin Passwordless Login 5

Description 5

Risk 5

Recommendation 5

Finding 3 – rexec Service Running 6

Description 6

Risk 6

Recommendation 6

Finding 4 – Operating System (OS) End of Life (EOL) Detection 6

Description 6

Risk 6

Recommendation 6

Finding 5 - Distributed Ruby (dRuby/DRb) Multiple Remote Code Execution Vulnerabilities 7

Description 7

Risk 7

Recommendation 7

Finding 6 - Possible Backdoor: ingreslock 7

Description 7

Risk 7
Recommendation 8
Risk Assessment 8
Recommendations 8
High Priority 8
Medium Priority 8
Long-Term Improvements 8
Conclusion 9
Appendix A 9
Appendix B 9

Vulnerability Assessment Report

Project: Vulnerability Assessment of Metasploitable 2 Using Greenbone/OpenVAS

Author: Thato Montjane

Date: August 2026

Classification: Portfolio Project

Executive Summary

Purpose

The purpose of this assessment was to identify security vulnerabilities within a deliberately vulnerable Linux host (Metasploitable 2) using Greenbone/OpenVAS. The assessment aimed to evaluate the security posture of the target system and provide recommendations to reduce the identified risks.

Scope

The assessment included:

- One target host (Metasploitable 2)

- IP Address: **192.168.56.101**
- Vulnerability scanning using Greenbone/OpenVAS
- Risk analysis based on CVSS scoring

No exploitation of vulnerabilities was performed.

Assessment Objectives

The objectives of this assessment were to:

- Identify known vulnerabilities affecting the target.
- Classify vulnerabilities according to severity.
- Assess the overall security risk.
- Recommend remediation measures.
- Demonstrate practical vulnerability management techniques.

Lab Environment

Component	Details
Scanner	Greenbone/OpenVAS
Scanner Host	Kali Linux 2025.4
Target	Metasploitable 2
Virtualization Platform	Oracle VirtualBox
Target IP	192.168.56.101
Scan Profile	Full and Fast

Methodology

The assessment followed a structured vulnerability management process:

Phase 1 – Environment Setup

- Configured Kali Linux.
- Installed Greenbone/OpenVAS.
- Configured networking within VirtualBox.
- Verified communication between Kali Linux and Metasploitable.

Phase 2 – Target Configuration

- Created a target within Greenbone.

- Configured the Alive Test.
- Selected the Full and Fast scan profile.

Phase 3 – Vulnerability Scanning

Greenbone performed automated security testing to identify:

- Open ports
- Vulnerable services
- Outdated software
- Remote code execution vulnerabilities
- Authentication weaknesses

Phase 4 – Risk Assessment

Each finding was evaluated using:

- CVSS Base Score
- Severity Rating
- Potential Business Impact

Executive Findings

Severity	Findings
Critical	14
High	8
Medium	40
Low	6
Log	88

Overall Risk Rating:

🔴 **Critical (CVSS 10.0)**

The assessment identified a total of 68 vulnerabilities, including 14 Critical, 8 High, 40 Medium, and 6 Low severity findings. The presence of multiple CVSS 10.0 vulnerabilities indicates that the target system is at a critical level of risk and would require immediate remediation before being deployed in a production environment.

Critical Findings

Finding 1 – TWiki Remote Command Execution

Severity: Critical

CVSS: 10.0

Affected Port: 80/TCP

Description

Greenbone identified an outdated TWiki installation vulnerable to multiple security issues, including cross-site scripting (XSS) and remote command execution.

Risk

Successful exploitation could allow an attacker to execute arbitrary commands on the target system, potentially resulting in complete system compromise.

Recommendation

- Upgrade TWiki to a supported version.
- Apply vendor security patches.
- Remove unnecessary web applications.

Finding 2 – rlogin Passwordless Login

Severity: Critical

Affected Port: 513/TCP

Description

The rlogin service allows insecure remote login functionality.

Risk

An attacker may gain unauthorized access through weak or missing authentication controls.

Recommendation

- Disable rlogin.
- Replace with SSH.
- Restrict remote administrative access.

Finding 3 – rexec Service Running

Severity: Critical

Affected Port: 512/TCP

Description

The rexec service permits remote command execution while transmitting credentials in plaintext.

Risk

Attackers may intercept credentials or execute commands remotely.

Recommendation

- Disable the rexec service.
- Replace with encrypted remote management protocols.

Finding 4 – Operating System (OS) End of Life (EOL) Detection

- Severity: Critical (CVSS 10.0)
- Affected Port: 192.168.56.101

Description

The target system is running an operating system that has reached End of Life (EOL). This means it no longer receives security patches or vendor support.

Risk

Unsupported operating systems are highly vulnerable because newly discovered vulnerabilities are unlikely to be fixed. Attackers can exploit known weaknesses that remain permanently unpatched.

Recommendation

- Upgrade to a supported operating system.
- Apply the latest security updates.
- Establish an operating system lifecycle management process.

Finding 5 - Distributed Ruby (dRuby/DRb) Multiple Remote Code Execution Vulnerabilities

- **Severity:** Critical (CVSS 10.0)
- **Affected Port:** 8787/TCP

Description

The Distributed Ruby service (dRuby/DRb) contains multiple vulnerabilities that may allow remote code execution.

Risk

An attacker could execute arbitrary code remotely, potentially gaining full control of the system.

Recommendation

- Update or remove the vulnerable DRb service.
- Restrict network exposure.
- Disable the service if it is not required.

Finding 6 - Possible Backdoor: ingreslock

- **Severity:** Critical (CVSS 10.0)
- **Affected Port:** 1524/TCP

Description

Greenbone detected the ingreslock service, which is commonly associated with backdoors in Metasploitable 2.

Risk

A backdoor provides attackers with unauthorized remote access to the system, bypassing normal authentication mechanisms.

Recommendation

- Remove unauthorized backdoors immediately.
- Investigate the system for signs of compromise.
- Reinstall or rebuild affected systems if necessary.

Risk Assessment

The assessment identified multiple critical services that significantly increase the attack surface of the target system. Legacy protocols and outdated software expose the host to remote compromise and unauthorized access.

The presence of several critical vulnerabilities indicates that the system would require immediate remediation before deployment in a production environment.

Recommendations

High Priority

- Remove insecure legacy services.
- Disable rexec.
- Disable rlogin.
- Update vulnerable web applications.
- Apply available security patches.

Medium Priority

- Reduce unnecessary open ports.
- Conduct regular vulnerability assessments.
- Implement secure configuration baselines.

Long-Term Improvements

- Introduce continuous vulnerability management.
- Perform periodic penetration testing.
- Strengthen network segmentation.
- Monitor systems using centralized logging and security monitoring tools.

Conclusion

The assessment successfully identified several critical security weaknesses affecting the target system. Greenbone/OpenVAS provided detailed visibility into vulnerable services and outdated software, enabling effective risk prioritization and remediation planning.

Although Metasploitable 2 is intentionally vulnerable for educational purposes, the assessment demonstrates the importance of regular vulnerability management and proactive security practices in protecting organizational systems.

Appendix A

Tools Used

- Kali Linux 2025.4
- Greenbone/OpenVAS
- Oracle VirtualBox
- Metasploitable 2

Target Information

- Host: Metasploitable2
- IP Address: 192.168.56.101
- Scan Configuration: Full and Fast

Appendix B

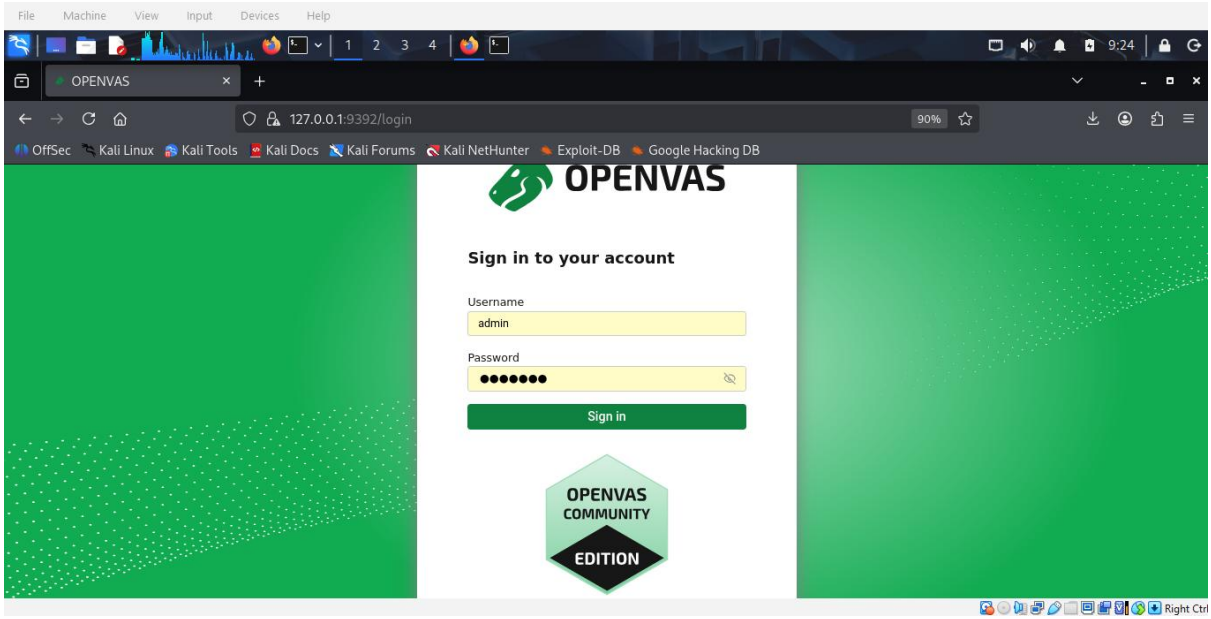


Figure 1-Greenbone login

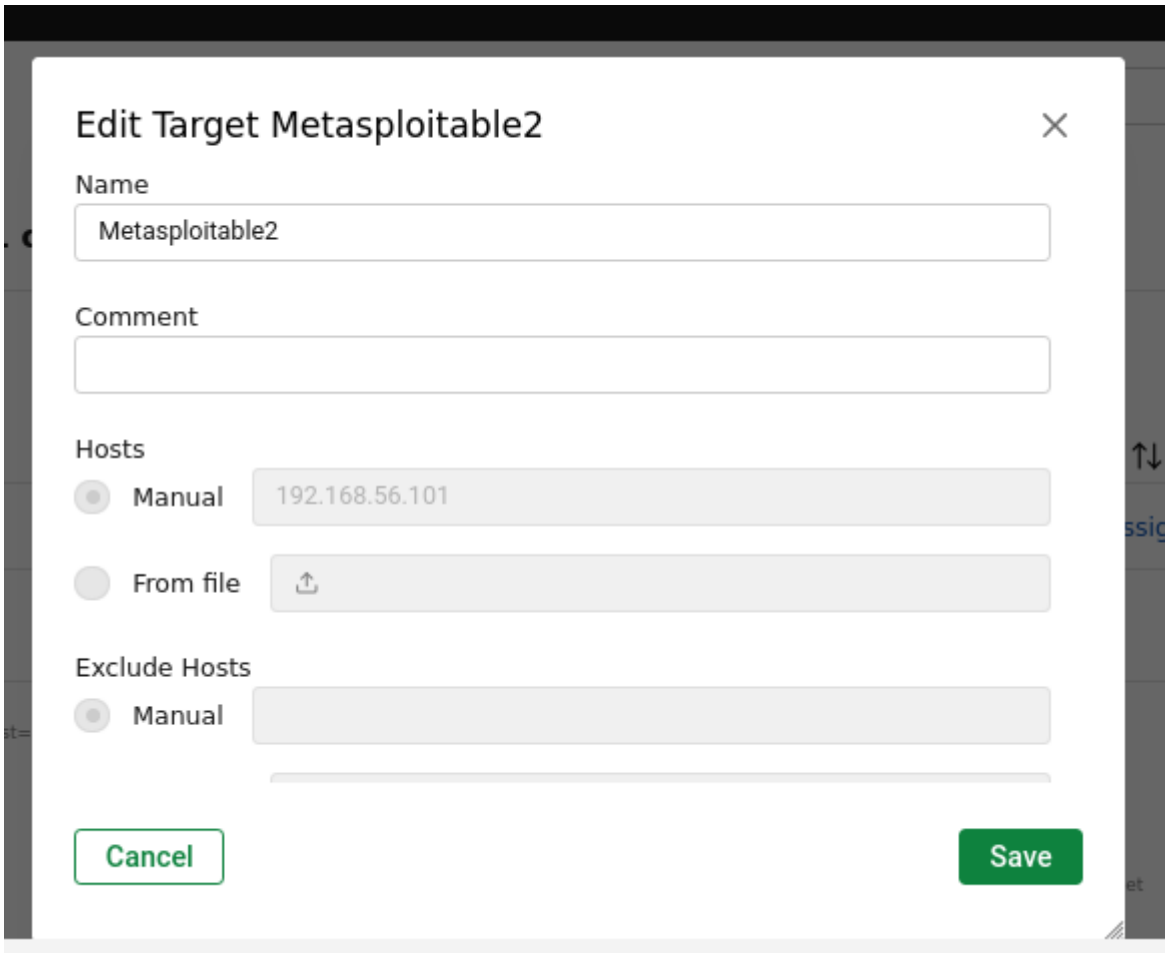


Figure 2-Task Configuration

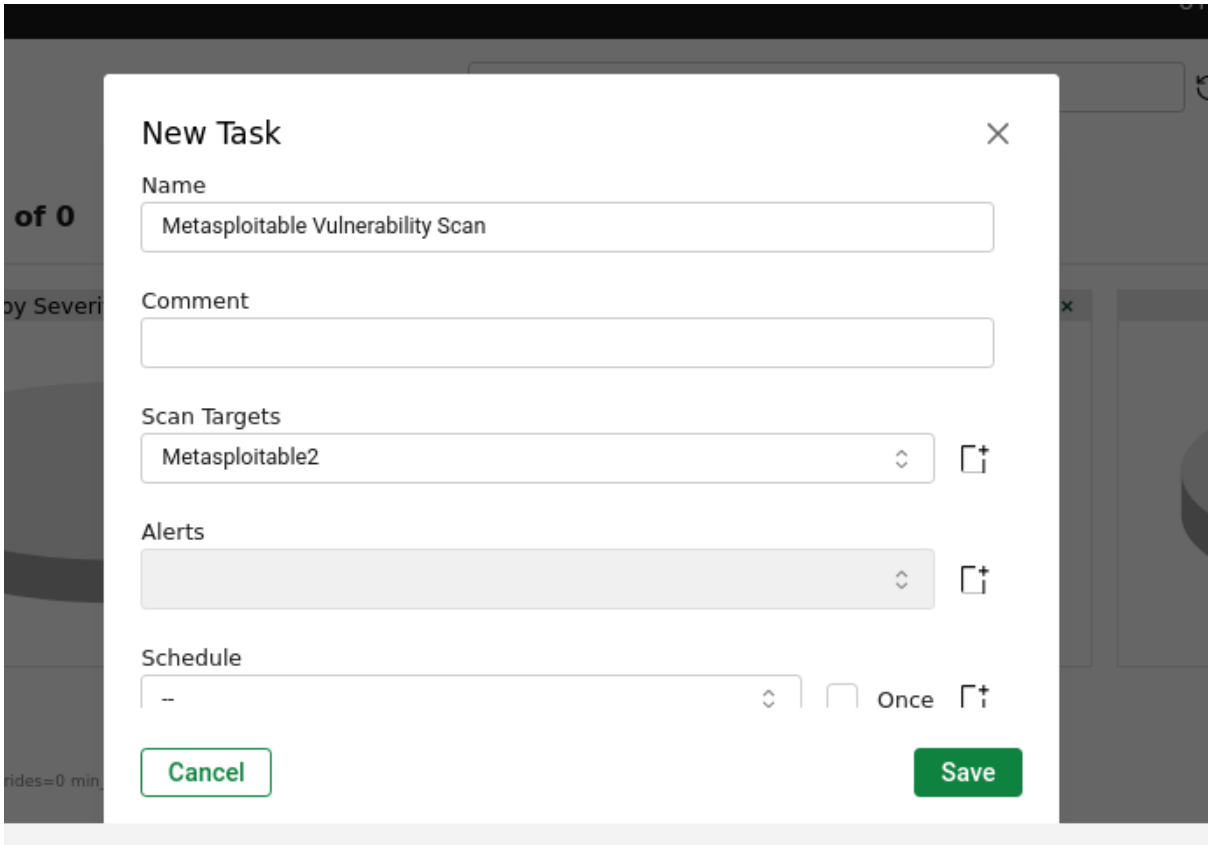


Figure 3-Scan Configuration

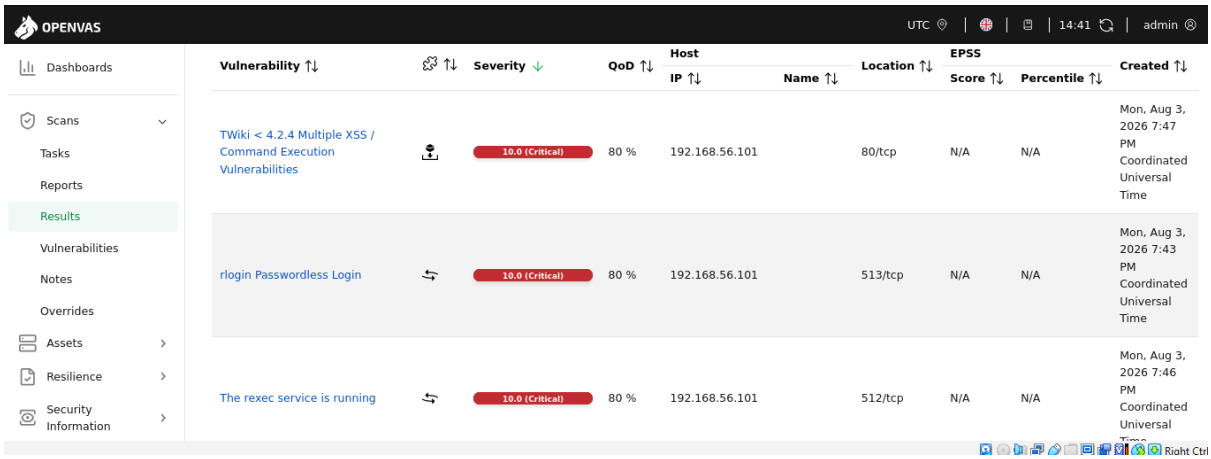


Figure 4-Critical Findings

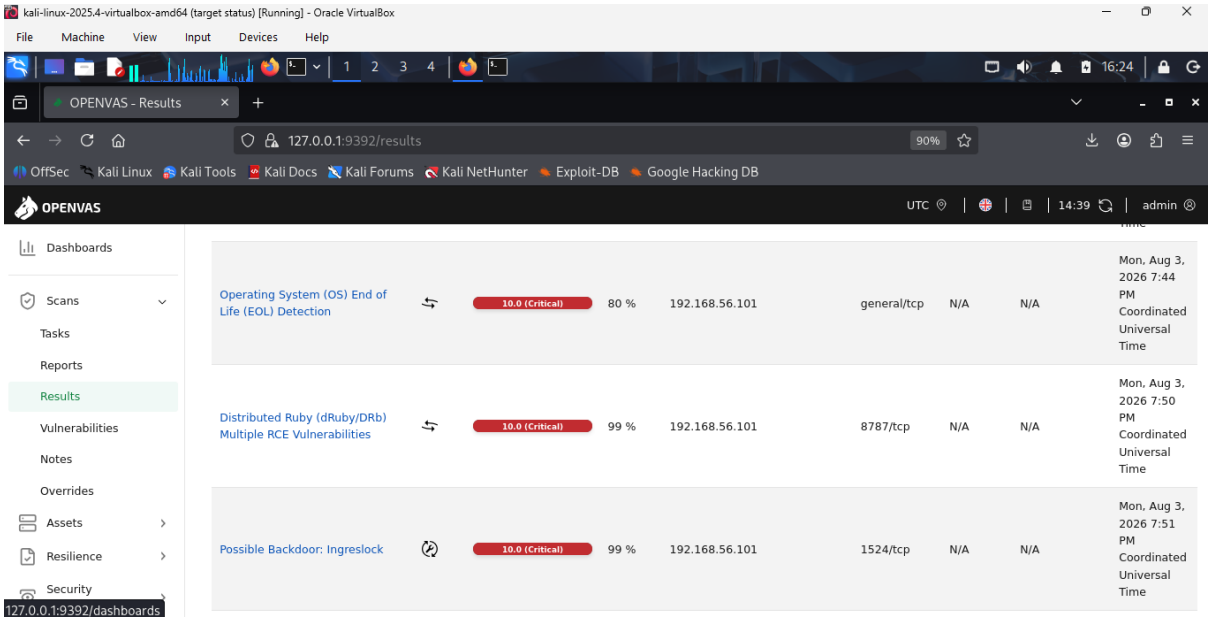


Figure 5-Critical Findings

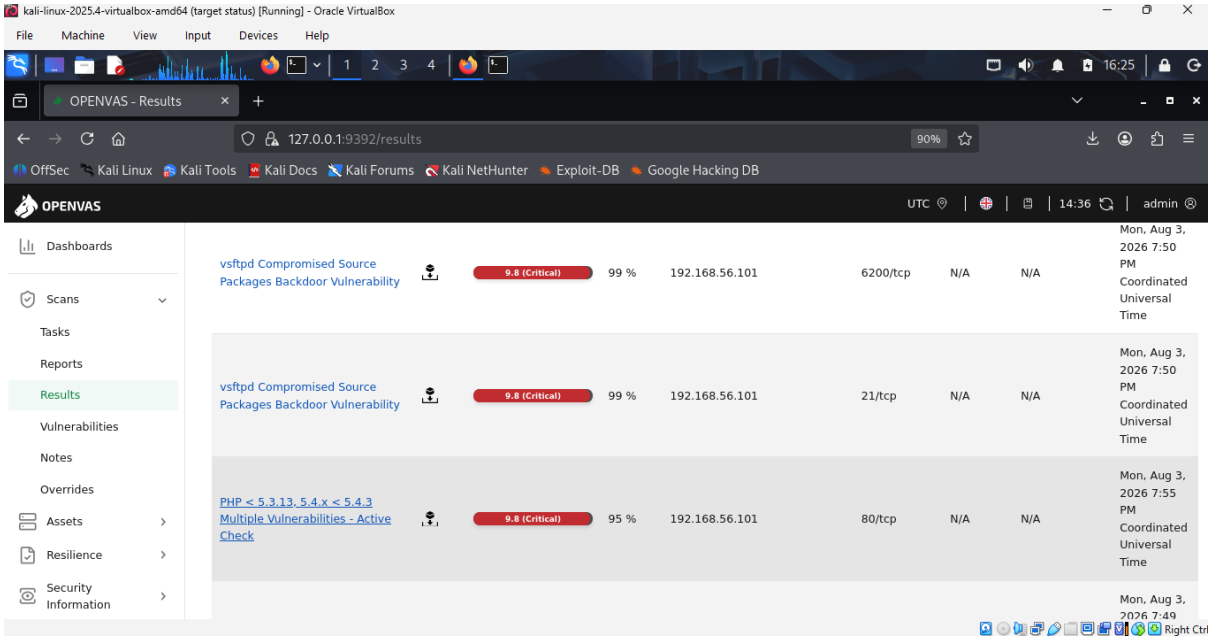


Figure 6-Critical Findings

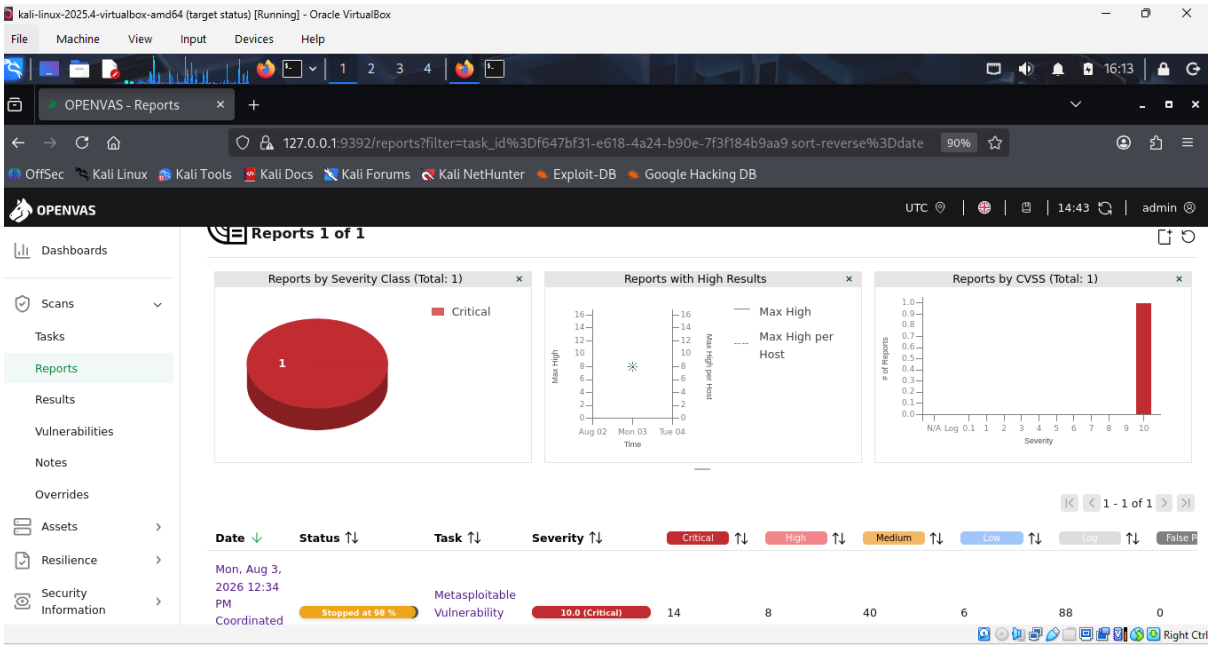


Figure 7-Report Summary

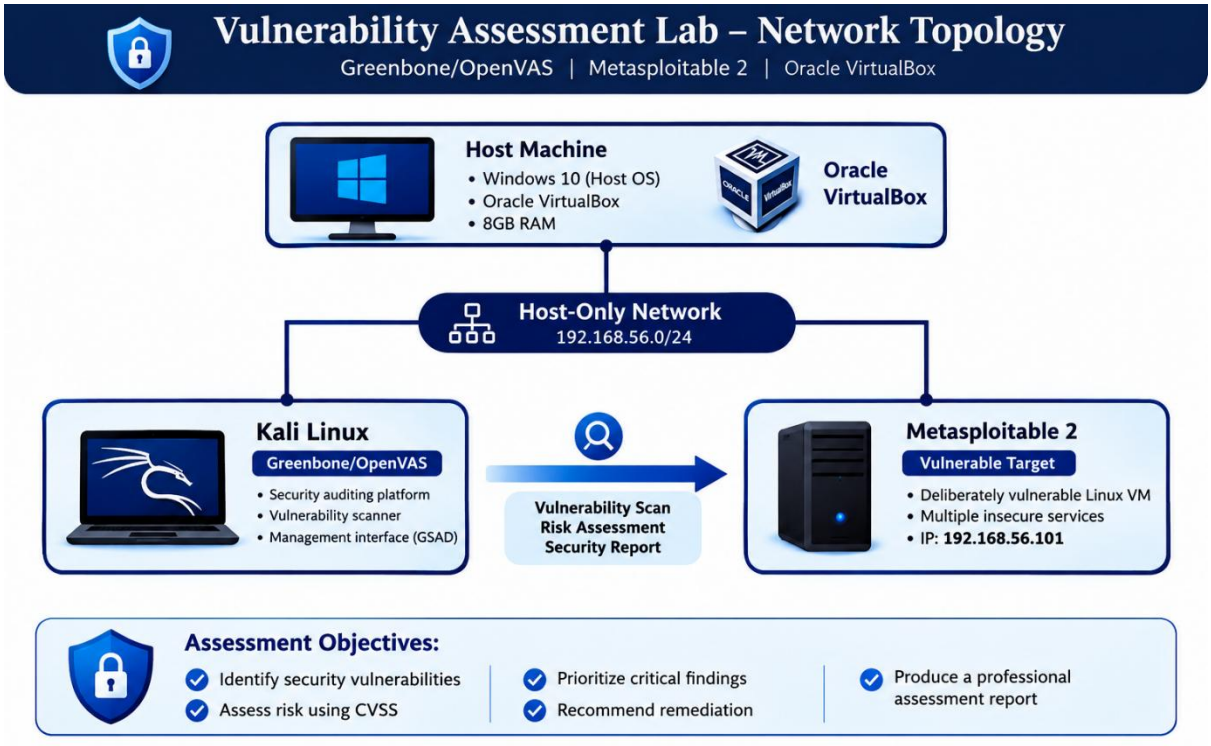


Figure 8- Network Topology